

LESSON 3-7 • SESSION 1 • PRACTICE • FORM A

Customary vs. Metric Conversions

Learning target – I can convert a measurement between the customary and metric systems using a given rate. • 6.RP.A.3d

REMEMBER

- Use the fact the problem gives. $\approx$ means “about equal to,” so answers are **estimates**.
- With 1 kg $\approx$ 2.2 lb: kg to lb, **multiply** ($10 \times 2.2 = 22$ lb). Lb to kg, **divide**.
- Before you compare two amounts, put both in the **same unit**.

WORKED EXAMPLE

A bag weighs 44 pounds. About how many kilograms? (1 kg $\approx$ 2.2 lb)

Step 1 A kilogram is heavier, so the number gets smaller: divide.

Step 2 $44 \div 2.2 = 20$.

Step 3 Check: $20 \times 2.2 = 44$.

Answer 44 lb $\approx$ 20 kg

WORDS conversion factor (*factor de conversión*) • approximately ($\approx$) (*aproximadamente*) • reasonableness (*razonabilidad*)

PRACTICE

show your work • label units

1. Convert. (1 kg $\approx$ 2.2 lb, 1 m $\approx$ 3.28 ft, 1 in = 2.54 cm)

a. 10 kg $\approx$ ____ lb

b. 44 lb $\approx$ ____ kg

c. 5 m $\approx$ ____ ft

d. 4 in = ____ cm

2. **Speed Sign.** A highway sign in Canada reads 100 km per hour. (1 mi $\approx$ 1.6 km)

Hint A mile is longer than a kilometer, so divide.

a. About how many miles per hour is that? _____ mph

b. Should your answer be greater or less than 100? Why? _____

Show your work.

3. **Soccer Field.** A soccer field is 70 m wide. A football field is 160 ft wide. (1 m $\approx$ 3.28 ft)

a. Soccer width in feet: _____ ft

b. Which field is wider? _____

Show your work.

4. **Suitcase.** The bag limit is 50 lb. A suitcase weighs 20 kg. (1 kg $\approx$ 2.2 lb)

a. Weight in pounds: _____ lb

b. Under the limit? By how much? _____

Show your work.

5. **Road Trip.** A drive to Montreal is 150 miles. About how many kilometers is that? (1 mi $\approx$ 1.6 km)

Hint Kilometers are shorter, so the number gets bigger: multiply.

Show your work.

Answer _____ km

- 6. Water Bottle.** A bottle holds 16.9 fluid ounces. About how many milliliters is that, to the nearest whole mL? (1 fl oz $\approx$ 29.6 mL)

Show your work.

Answer _____ mL

- 7. Backpack Check.** A backpack weighs 7 kg. The nurse says it should stay under 16 pounds. Find its weight in pounds. Is it under the limit? (1 kg $\approx$ 2.2 lb)

Hint Change kilograms to pounds, then compare with 16.

Show your work.

Answer _____ lb

- 8. Which Is Longer?** A rope is 4 m long. A cable is 12 ft long. (1 m $\approx$ 3.28 ft)

Hint Put both lengths in feet before you compare.

- a.** Rope length in feet: _____ ft
b. Which is longer, and by about how much? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Kelly's Work.** Kelly converted 50 miles to kilometers by dividing: $50 \div 1.6 = 31.25$ km. Explain why 31 km is impossible for 50 miles. Then find the correct distance. (1 mi $\approx$ 1.6 km)

TEST PRACTICE

choose the best answer

- 10.** 2 meters is about how many feet? (1 m $\approx$ 3.28 ft)

A 1.64 **B** 6.56 **C** 5.28 **D** 32.8

- 11.** 4.4 pounds is about how many kilograms? (1 kg $\approx$ 2.2 lb)

A 9.68 **B** 6.6 **C** 2.2 **D** 2

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

44 R	9.09 X	15.4 C	16.4 M	93.75 Q	229.6 T	62.5 E	240 I	160 Z
		#1c	#2a	#3a	#4a	#5	#7	

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____